

Fundamentals of Probability

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Experiment:- Experiment is an act that can be repeated under given conditions.

Ex- Tossing a coin, throwing a dice etc.

Outcomes:- The result of an experiment are called outcomes.

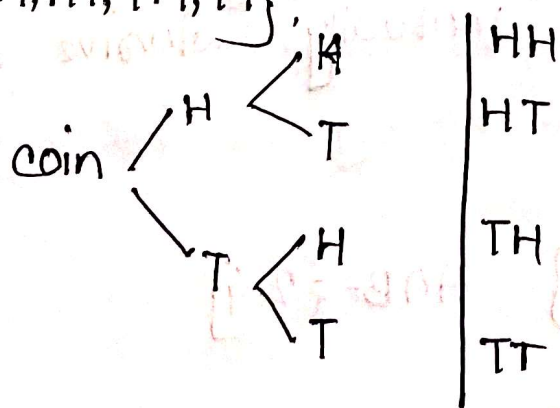
Random experiment:- Experiment are called random experiments if the outcomes depend on chances and cannot be predicted with certainty.

Ex- Tossing a fair coin, throwing of dice etc.

Sample Space:- The collection or totality of all possible outcomes of a random experiment is called sample space. It is denoted by S or Ω

Example:- 1. If we toss a coin the sample space is $S = \{H, T\}$.
2. If we toss a coin two times, the sample space is.

$S = \{HH, HT, TH, TT\}$



Each element of a sample space is called sample point.

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Event: Any subset of a sample space is called event.

It is denoted by A, B, C . There are two types of event.

They are:

1. Simple Event.

2. Compound Event.

Mutually Exclusive Event: Two events are said to be mutually exclusive if they have no common points.

Ex - $A = \{3, 5, 1\}$ $B = \{2, 4, 6\}$ then $A \cap B = \emptyset$

$$AB = \emptyset \quad P(AB) = P(A \cap B) = 0.$$

If two events are mutually exclusive, then their probability is 0.

Non-mutually Exclusive Event: Two events are said to be not mutually exclusive event if they have common points.

If A and B are not mutually exclusive events, then

$$AB \neq \emptyset$$

$$A = \{1, 2, 4\} \quad B = \{2, 4, 6\} \quad A \cap B = \{2, 4\}$$

Conditional Probability:- If A and B are two events in S. Then the conditional probability of A for given value of B denoted by $P[A|B]$ is defined by

$$P(A|B) = \frac{P(AB)}{P(B)} ; P(B) > 0$$

Similarly, $\rightarrow$ A conditional B / A given B, B ঘটনা আগে ঘটে, B কে base করে A নির্ণয়

$$P(B|A) = \frac{P(AB)}{P(A)} ; P(A) > 0$$

$\rightarrow$ B conditional A / B given A, A ঘটনা আগে ঘটে, A কে base করে B নির্ণয়

Independent Event:- Two events A and B are said to be independent, if and only if one of the following conditions holds:

$$i) P[AB] = P[A]P[B]$$

$$ii) P[A|B] = P[A]$$

$$iii) P[B|A] = P[B]$$

Dependent Event:- Two events A and B are said to be dependent if and only if one of the following conditions holds:

$$i) P[AB] \neq P[A]P[B]$$

$$ii) P[A|B] \neq P[A]$$

$$iii) P[B|A] \neq P[B]$$

Definition of Probability:-Classical or Mathematical Probability (Priori Probability):-

If A is the event in S then the probability of the event A is denoted by $P(A)$ is defined as

$$P(A) = \frac{\text{favourable number of outcomes}}{\text{total number of outcomes}}$$

$$P(A) = \frac{N(A)}{N(S)} = \frac{m}{n}; \quad 0 \leq P(A) \leq 1$$

Example:-

$$S = \{1, 2, 3, 4, 5, 6\} \quad A = \{1, 2\}$$

$$N(S) = 6$$

$$N(A) = 2$$

$$\therefore P(A) = \frac{N(A)}{N(S)} = \frac{2}{6} = \frac{1}{3} \quad (\text{Ans})$$

Axiomatic Probability:-

Suppose S is a sample space and A is an event of this sample space. Then the probability of the event A denoted by $P(A)$ must satisfy the following axioms:

- i. $P(A) \geq 0$
- ii. $P(S) = 1$
- iii. If A and B are mutually exclusive events.
- iv. $P(A_1 \cup A_2 \cup \dots \cup A_k) = P(A_1) + P(A_2) + \dots + P(A_k)$

Additive laws of Probability:

1. Mutually exclusive event
2. Non mutually exclusive event

If A and B are two events then,

$$P[A \cup B] = P[A] + P[B] \longrightarrow \text{mutually exclusive event}$$

$$P[A \cup B] = P[A] + P[B] - P[AB] \longrightarrow \text{non mutually exclusive event}$$

Theorem-1

The values of probability lies between 0 to 1.
i.e. $0 \leq P(A) \leq 1$

Theorem-2

$$P(A) + P(\bar{A}) = 1$$

1. What are the different approaches of defining probability?

There are 4 approaches of defining probability.

They are —

1. Classical or mathematical definition of probability.
2. Empirical or statistical Probability.
3. Subjective Probability
4. Axiomatic Probability.

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2. Difference between experiment and random experiment

Experiment	Random Experiment
1. Experiment is an act that can be repeated under given conditions.	1. Experiment is called random experiment if its outcomes depends on chances and can't be predicted with certainty.
2. Its outcomes can be biased.	2. Its outcomes can't be biased.
3. An experiment can have any number of possible outcomes.	3. A random experiment has more than one possible outcomes.
4. Tossing with a biased coin.	4. Tossing a fair coin. (H) (T)

3. Difference between independent event and dependent event

Independent Event	Dependent Event
1. When probability of one event does not affect the probability of another event, they are known as independent events.	1. The probability of one event does affect the probability of another event, then they are called dependent events.

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4. Difference between mutually exclusive and non mutually exclusive event.

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5. Discuss classical probability and axiomatic probability

Ques-

In a factory, machines A, B, and C manufacture respectively 25%, 35%, 40% of their product of the total of their output 5, 4 and 2 percent are defective bolts respectively. A bolt is drawn at random from the product and is found to be defective. What are the probabilities of that it was manufactured by machine (i) A and (ii) C.

Let,

B_1 item is produced by machine A

B_2 " " " " " B

B_3 " " " " " C

D defective " " " " all machines.

We have,

$$P[B_1] = 25\% = 0.25$$

$$P[B_2] = 35\% = 0.35$$

$$P[B_3] = 40\% = 0.4$$

and,

$$P[D|B_1] = 5\% = 0.05$$

$$P[D|B_2] = 4\% = 0.04$$

$$P[D|B_3] = 2\% = 0.02$$

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We know,

$$P(B_i|D) = \frac{P(B_i) \cdot P(D|B_i)}{\sum_{i=1}^n P(B_i) \cdot P(D|B_i)}$$

$$\therefore P(B_1|D) = \frac{P(B_1) \cdot P(D|B_1)}{P(B_1) \cdot P(D|B_1) + P(B_2) \cdot P(D|B_2) + P(B_3) \cdot P(D|B_3)}$$

$$= \frac{0.25 \times 0.05}{0.25 \times 0.05 + 0.35 \times 0.04 + 0.4 \times 0.02}$$

$$= \frac{0.0125}{0.0345}$$

$$= 0.323. \text{ (Ans.)}$$

$$\therefore P(B_3|D) = \frac{P(B_3) \cdot P(D|B_3)}{\sum P(B_i) \cdot P(D|B_i)}$$

$$= \frac{0.4 \times 0.02}{0.0345}$$

$$= 0.232. \text{ (Ans.)}$$

State Additive law of Probability and multiplicative law of probability.

Non mutually exclusive event,

$$P(A \cup B) = P(A) + P(B) - P(AB)$$

Mutually exclusive event,

$$P(A \cup B) = P(A) + P(B)$$

If A, B and C are events then,

Non mutually exclusive event,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(BC) - P(CA) + P(ABC)$$

Mutually exclusive event,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

Multiplicative law:-

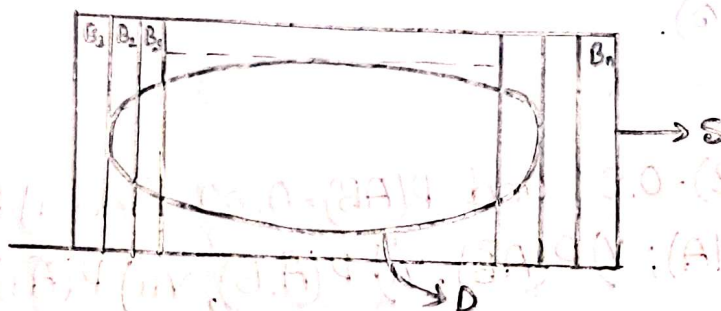
$$\begin{aligned} P(AB) &= P(A|B) \cdot P(B) \\ &= P(B|A) \cdot P(A) \end{aligned}$$

State Bayes theorem.

Let $B_1, B_2, B_3, \dots, B_n$ be a sequence of mutually exclusive events D be the event in S .

Then,

$$P(B_i|D) = \frac{P(B_i|D) \cdot P(B_i)}{\sum_{i=1}^n P(B_i) \cdot P(D|B_i)} ; i = 1, 2, 3, \dots, n.$$



Prblm-1

A and B are two mutually exclusive events with $P[A] = 0.35$ and $P[B] = 0.15$. Find $P[A \cup B]$, $P[\bar{A}]$, $P(A \cap B)$, $P(\bar{A} \cup \bar{B})$.

Given,

$$P[A] = 0.35,$$

$$P[B] = 0.15$$

$$\begin{aligned} \therefore P[A \cup B] &= P[A] + P[B] \\ &= 0.35 + 0.15 \\ &= 0.5, \end{aligned}$$

$$\begin{aligned} \therefore P(\bar{A}) &= 1 - P[A] \\ &= 1 - 0.35 \\ &= 0.65. \end{aligned}$$

$$\therefore P(A \cap B) = 0$$

Since, A and B are mutually exclusive.

$$\therefore P(\bar{A} \cup \bar{B}) = P(\overline{A \cap B})$$

$$= 1 - P(A \cap B)$$

$$= 1 - 0$$

$$= 1. \text{ (Ans.)}$$

Prblm-2

If $P(A) = 0.6$, $P(B) = 0.8$ and $P(AB) = 0.50$. Find i) $P(\bar{A})$, ii) $P(A \cup B)$, iii) $P(A|B)$, iv) $P(B|A)$; v) $P(A\bar{B})$, vi) $P(\bar{A} \cdot B)$, vii) $P(\bar{A}\bar{B})$, viii) $P(\bar{A} \cup \bar{B})$
 xi) Are the events A and B independent? x) Are A and B mutually exclusive?

Given that,

$$P(A) = 0.6$$

$$P(B) = 0.8$$

$$P(AB) = 0.50$$

i. we know,

$$P(A) + P(\bar{A}) = 1$$

$$\therefore P(\bar{A}) = 1 - P(A)$$

$$= 1 - 0.6$$

$$= 0.4. \text{ (Ans.)}$$

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ii) By additive law of probability

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= 0.6 + 0.8 - 0.50$$

$$= 0.9 \text{ (Ans)}$$

$$\text{iii) } P(A|B) = \frac{P(AB)}{P(B)} \quad [P(B) > 0]$$

$$= \frac{0.50}{0.8}$$

$$= 0.625 \text{ (Ans)}$$

$$\text{iv) } P(B|A) = \frac{P(AB)}{P(A)}$$

$$= \frac{0.50}{0.6}$$

$$= 0.833 \text{ (Ans)}$$

$$\text{v) } P(A\bar{B}) = P(A \cap \bar{B})$$

$$= P(A) - P(AB)$$

$$= 0.6 - 0.50$$

$$= 0.10 \text{ (Ans)}$$

$$\text{vi) } P(\bar{A}B) = P(B) - P(AB)$$

$$= 0.8 - 0.50$$

$$= 0.30 \text{ (Ans)}$$

$$\text{vii) } P(\bar{A}\bar{B}) = P(\bar{A} \cap \bar{B})$$

$$= 1 - P(A \cup B)$$

$$= P(\overline{A \cup B})$$

$$= 1 - P(A \cup B)$$

$$= 1 - 0.9$$

$$= 0.1 \text{ (Ans)}$$

$$\text{viii) } P(\overline{A \cup B}) = 1 - P(A \cup B)$$

$$= 1 - 0.9$$

$$= 0.1$$

$$\text{ (Ans) }$$

ix) The events A and B are independent if

$$P[AB] = P[A]P[B]$$

Here,

$$P[AB] = 0.5.$$

$$\therefore P[A]P[B] = 0.6 \times 0.8 \\ = 0.48.$$

Ans, $P(AB) \neq P(A)P(B)$, the events are not independent.
(Ans)

x) A and B are mutually exclusive if $P[AB] = 0$

But,

$$\text{here } P(AB) = 0.5$$

which isn't equal 0,

So, A and B are not mutually exclusive event.

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Problem-3

A and B are two mutually exclusive events with $P[A] = 0.35$ and $P[B] = 0.15$. Find

i) $P[A \cup B]$

Given that,

$$P[A] = 0.35$$

$$P[B] = 0.15$$

$$\begin{aligned} \therefore P[A \cup B] &= P[A] + P[B] \\ &= 0.35 + 0.15 \\ &= 0.5. \quad (\text{Ans}) \end{aligned}$$

ii) $P(\bar{A})$.

We know,

$$P(A) + P(\bar{A}) = 1.$$

$$\begin{aligned} \Rightarrow P(\bar{A}) &= 1 - P(A) \\ &= 1 - 0.35 \\ &= 0.65. \quad (\text{Ans}) \end{aligned}$$

iii) $P(A \cap B)$.

$$P(A \cap B) = 0. \quad (\text{Ans})$$

Ans they are mutually exclusive events.

$$\begin{aligned} \text{iv) } P(\bar{A} \cap \bar{B}) &= P(\overline{A \cap B}) \\ &= 1 - P(A \cap B) \\ &= 1 - 0 \\ &= 1. \quad (\text{Ans}) \end{aligned}$$

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Problem-4

If $P(A) = \frac{1}{3}$, $P(B) = \frac{3}{4}$, $P(A \cup B) = \frac{11}{12}$. Find i) $P(A|B)$ ii) $P(\bar{A}|B)$

Given that,

$$P(A) = \frac{1}{3}$$

$$P(B) = \frac{3}{4}$$

$$P(A \cup B) = \frac{11}{12}$$

We know,

$$P(A \cup B) = P(A) + P(B) - P(AB)$$

$$\Rightarrow \frac{11}{12} = \frac{1}{3} + \frac{3}{4} - P(AB)$$

$$\Rightarrow P(AB) = \frac{11}{12} - \frac{1}{3} - \frac{3}{4}$$
$$= \frac{1}{6}$$

$$\text{i) } P(A|B) = \frac{P(AB)}{P(B)}$$
$$= \frac{1/6}{3/4}$$

$$= \frac{2}{9} \text{ (Ans)}$$

$$\text{ii) } P(\bar{A}|B) = \frac{P(\bar{A}B)}{P(B)}$$

$$= \frac{P(B) - P(AB)}{P(B)}$$

$$= \frac{3/4 - 1/6}{3/4} = \frac{7}{9} \text{ (Ans)}$$